

# Le système international d'unités The International System of Units

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Annexe 2

## Annex 2

Valeurs recommandées des fréquences étalons  
destinées à la mise en pratique de la définition du  
mètre et aux représentations secondaires de la  
seconde

**Recommended values of standard frequencies  
for applications including the practical  
realization of the metre and secondary  
representations of the second**

Partie 2

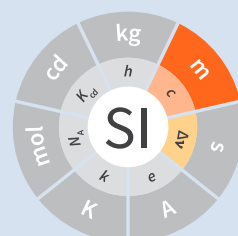
## Part 2

Mercury 199 Ion ()

**Mercury 199 Ion ()**

CCL-CCTF Frequency Standards Working Group

July 2013  
juillet 2013





**Recommended values of standard  
frequencies for applications including  
the practical realization of the metre and  
secondary representations of the second  
Part 2: Mercury 199 Ion ()**

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**9<sup>th</sup> edition      2019**

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## 1. Recommended value [1] of the frequency:

$$f(^{199}\text{Hg}^+) = 1\,064\,721\,609\,899\,145.30\text{ Hz}$$

equivalent to

$$\lambda(^{199}\text{Hg}^+) = 281\,568\,867.591\,968\,5\text{ fm}$$

with a relative uncertainty of  $1.9 \times 10^{-15}$ .

This radiation was already endorsed as a secondary representation of the definition of the second.

## 2. Source data

Adopted value  $f(^{199}\text{Hg}^+) = 1\,064\,721\,609\,899\,145.30\text{ Hz}$   
 $u_c/y = 1.9 \times 10^{-15}$

calculated from

| $f(^{199}\text{Hg}^+)/\text{Hz}$ | $u/\text{Hz}$ | source data |
|----------------------------------|---------------|-------------|
| 1 064 721 609 899 145.30         | 0.69          | [2]         |

As this value was issued from only one laboratory the CCTF considered it prudent to estimate a relative standard uncertainty of  $1.9 \times 10^{-15}$ .

**$^{199}\text{Hg}^+$  ion,  $5d^{10}6s\ ^2S_{1/2} - 5d^96s^2\ ^2D_{5/2}$  unperturbed optical transition**

## References

- [1] CIPM Recommendation 1(CI-2013): Updates to the list of standard frequencies.
- [2] J.E. Stalnaker, S.A. Diddams, T.M. Fortier, K. Kim, L. Hollberg, J.C. Bergquist, W.M. Itano, M.J. Delany, L. Lorini, W.H. Oskay, T.P. Heavner, S.R. Jefferts, F. Levi, T.E. Parker and J. Shirley: Optical-to-microwave frequency comparison with fractional uncertainty of  $10^{-15}$ , *Appl. Phys. B*, **89**, 167 (2007).